# NeuralMed: A Refined Multi-Disease Detection System with Doctor Portal, Cross-Validated Models, and LLM Clinical Summaries

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***Abstract***—This paper documents Deliverable 3 of NeuralMed, a multi-disease prediction platform. Since Deliverable 2, significant improvements have been made across four dimensions: (1) model reliability — overfitting in tree-based models addressed through max_depth constraints and 5-fold stratified cross-validation, yielding honest accuracy estimates; (2) clinical tooling — a Doctor Portal added with per-prediction confidence scores, XGBoost/Random Forest feature importance bars, clinical reference ranges, and Gemini-powered clinical AI summaries; (3) user interface — complete dark-themed redesign replacing Materialize carousel with a modular CSS design system and an embedded AI chatbot; and (4) API stability — Gemini model updated from deprecated gemini-pro to gemini-2.5-flash-lite with retry logic. Updated evaluation across five diseases (diabetes, heart, Parkinson's, liver, breast cancer) shows test accuracies ranging 69–98% and cross-validation AUC values 0.77–0.99.

***Index Terms***—*multi-disease detection, deep learning, cross-validation, feature importance, clinical decision support, doctor portal, LLM, Flask, healthcare AI*

## I. PROJECT SUMMARY

NeuralMed addresses fragmented early-stage medical screening by providing a unified web platform where users input clinical data or medical images and receive a prediction alongside AI-generated health guidance. The system integrates classical ML, deep learning (VGG-16 transfer learning for kidney CT), and Google Gemini LLM within a single Flask application.

Since Deliverable 2, the system expanded from a patient-facing screening tool to a dual-portal platform. The new Doctor Portal provides clinical decision support with confidence scoring, model-attribution charts, and physician-targeted AI summaries — addressing the transparency gap identified in D2. Model evaluation was also made more rigorous by applying 5-fold stratified cross-validation and depth constraints to all tree-based models.

## II. UPDATED SYSTEM ARCHITECTURE AND PIPELINE

The pipeline retains the four-layer modular design from D2 but adds the Doctor Portal inference layer and updates the Interface Layer with the redesigned frontend. The end-to-end flow is:

Dataset → Preprocessing → Model → Flask Route → [Patient Template | Doctor Template + LLM Summary]

### *A. Data and Processing Layer*

Raw CSV datasets (UCI/Kaggle) are DVC-tracked. Preprocessing in D3 enforces a consistent 80/20 stratified split with random_state=42 across all diseases. SimpleImputer handles missing liver values; StandardScaler normalizes all tabular features.

### *B. Model Layer*

D3 imposes max_depth=10 on Random Forest/Extra Trees and max_depth=5 on XGBoost. Decision Tree is capped at max_depth=8. Logistic Regression uses max_iter=1000. The 5-fold StratifiedKFold strategy replaces single-split evaluation. Model artifacts (.pkl) are loaded lazily at inference time.

### *C. Doctor Portal Layer (new in D3)*

Each doctor route loads the preprocessor and model separately, calls predict_proba() for confidence, extracts feature_importances_ (XGB/RF) or |coef_| (LogisticRegression) for attribution, and calls a physician-targeted Gemini prompt. Results are passed to five new clinical templates.

### *D. Interface Layer*

The Flask application exposes patient routes (/diabetes, /heart, /liver, /parkinsons, /bcancer, /kidney) unchanged, and adds five doctor routes (/doctor/diabetes through /doctor/bcancer) plus an embedded /chat chatbot endpoint. Static assets use a unified main.css

design system replacing 8 separate stylesheets.

## III. REFINEMENTS MADE SINCE DELIVERABLE 2

### *A. Overfitting Mitigation*

D2 identified that RandomForest and DecisionTree achieved 100% test accuracy on the heart disease Kaggle dataset — a known artifact of label leakage in that particular dataset composition. D3 addresses this by: (1) applying max_depth constraints to all tree-based models; (2) evaluating all models with 5-fold stratified CV; and (3) reporting both test accuracy and CV mean±std so variance from dataset size is transparent.

### *B. Doctor Portal*

A new /doctor landing page lists five clinical assessment modules. Each module presents the same input form as the patient view but augmented with: (1) reference range hints displayed inline beneath each input field; (2) a confidence % bar color-coded green/amber/red; (3) horizontal feature importance bars sorted by contribution; (4) a patient value summary grid; (5) a Gemini clinical summary structured as four sections (Clinical Interpretation, Key Risk Factors, Recommended Next Steps, Important Caveats); (6) a browser-print PDF export via @media print CSS.

### *C. Gemini API Stability*

The deprecated gemini-pro and gemini-pro-vision model identifiers from D2 were replaced with gemini-2.5-flash-lite. The /chat endpoint adds 2-attempt retry logic with explicit quota-error detection, surfacing meaningful fallback messages rather than silent failures.

### *D. UI Redesign and FOUC Fix*

All 15 original templates were rewritten using a unified design system (main.css) with CSS custom properties (--bg, --primary, --grad, etc.). The Materialize carousel services page was replaced with a card-grid layout. A Flash of Unstyled Content (FOUC) was eliminated by setting background on the html element, ensuring the dark theme paints before the stylesheet fully loads.

### *E. Embedded AI Chatbot*

The static /chatbot route previously launched a separate Streamlit process. D3 replaces it with an embedded widget on the landing page: a fixed-action-button opens a sliding glass panel connected to a new /chat POST endpoint. The bot uses a NeuralMed-aware system prompt, responds in under 180 words, and handles Gemini rate limits gracefully.

## IV. INTERFACE USABILITY AND IMPROVEMENTS

Interface improvements were validated through functional testing of the golden path (form submission → prediction → result display) and edge cases (empty fields, API quota exceeded, model file missing). Key changes:

- Landing page: hero section with animated gradient orb, 4 statistics, 6 feature cards, disease module grid, About and Contact sections, footer, and embedded chatbot.
- Patient forms: 2-column form grids with unit labels, reference hints, always-visible result card with conditional styling (positive=red, negative=green).
- Doctor forms: same inputs augmented with inline reference ranges, clinical confidence bar, feature importance visualization, clinical AI summary with 4 structured sections, and print-to-PDF.
- Navigation: fixed navbar present on all pages; doctor/patient portals are distinct paths, clearly labeled.

The redesign reduces page-level CSS from 8 separate stylesheets to 1 unified file (659 lines) with semantic variables, cutting duplication from ~9,800 lines to ~659 lines.

## V. EXTENDED EVALUATION AND UPDATED RESULTS

### *A. Methodology Changes from D2*

D2 used a single 80/20 train-test split. D3 adds StratifiedKFold(5) cross-validation as the primary evaluation method and reports test accuracy, precision, recall, F1 (weighted), and CV mean±std for 8 models per disease.

### *B. Diabetes*

Table — Diabetes Model Comparison (D3)

| Model | Acc | Prec | Rec | F1 | CV Mean | CV Std |
|---|---|---|---|---|---|---|
| Naive Bayes | 70.8% | 71.8% | 70.8% | 71.1% | 75.5% | ±3.4% |
| Logistic Reg. | 71.4% | 70.6% | 71.4% | 70.8% | 77.5% | ±1.5% |
| KNN | 70.8% | 70.2% | 70.8% | 70.4% | 72.9% | ±2.4% |
| SVM | 74.7% | 74.4% | 74.7% | 74.5% | 76.0% | ±1.7% |
| Decision Tree | 74.7% | 73.9% | 74.7% | 73.7% | 68.1% | ±5.2% |
| Random Forest | 74.7% | 74.4% | 74.7% | 74.5% | 76.9% | ±2.3% |
| XGBoost | 73.4% | 73.1% | 73.4% | 73.2% | 73.2% | ±2.2% |
| Extra Trees | 72.1% | 71.2% | 72.1% | 71.2% | 77.1% | ±2.3% |

### *B. Heart Disease*

Table — Heart Disease Model Comparison (D3)

| Model | Acc | Prec | Rec | F1 | CV Mean | CV Std |
|---|---|---|---|---|---|---|
| Naive Bayes | 85.2% | 85.5% | 85.2% | 85.2% | 84.1% | ±4.3% |
| Logistic Reg. | 85.2% | 86.2% | 85.2% | 85.2% | 83.7% | ±4.9% |
| KNN | 79.6% | 81.1% | 79.6% | 79.7% | 81.5% | ±3.3% |
| SVM | 81.5% | 81.8% | 81.5% | 81.5% | 83.0% | ±5.9% |
| Decision Tree | 79.6% | 80.2% | 79.6% | 79.7% | 72.2% | ±6.4% |
| Random Forest | 83.3% | 83.4% | 83.3% | 83.4% | 80.7% | ±8.7% |
| XGBoost | 83.3% | 83.4% | 83.3% | 83.4% | 78.1% | ±6.9% |
| Extra Trees | 79.6% | 80.2% | 79.6% | 79.7% | 81.1% | ±5.2% |

### *B. Parkinson's*

Table — Parkinson's Model Comparison (D3)

| Model | Acc | Prec | Rec | F1 | CV Mean | CV Std |
|---|---|---|---|---|---|---|
| Naive Bayes | 56.4% | 78.3% | 56.4% | 58.1% | 59.5% | ±6.8% |
| Logistic Reg. | 82.1% | 81.1% | 82.1% | 80.9% | 83.1% | ±5.5% |
| KNN | 89.7% | 89.7% | 89.7% | 89.7% | 86.2% | ±2.6% |
| SVM | 87.2% | 89.1% | 87.2% | 85.6% | 84.1% | ±4.4% |
| Decision Tree | 82.1% | 84.4% | 82.1% | 82.7% | 87.7% | ±5.2% |
| Random Forest | 92.3% | 92.7% | 92.3% | 92.4% | 88.2% | ±4.8% |
| XGBoost | 89.7% | 90.8% | 89.7% | 90.0% | 88.7% | ±6.6% |
| Extra Trees | 89.7% | 89.7% | 89.7% | 89.7% | 89.2% | ±5.2% |

### *B. Liver Disease*

**Table — Liver Disease Model Comparison (D3)**

| Model | Acc | Prec | Rec | F1 | CV Mean | CV Std |
|---|---|---|---|---|---|---|
| Naive Bayes | 53.8% | 77.7% | 53.8% | 53.7% | 55.2% | ±4.1% |
| Logistic Reg. | 69.2% | 64.1% | 69.2% | 64.5% | 71.9% | ±3.4% |
| KNN | 63.2% | 63.6% | 63.2% | 63.4% | 67.1% | ±2.9% |
| SVM | 69.2% | 50.0% | 69.2% | 58.0% | 71.0% | ±0.5% |
| Decision Tree | 59.0% | 57.5% | 59.0% | 58.2% | 65.2% | ±1.1% |
| Random Forest | 62.4% | 58.8% | 62.4% | 60.2% | 69.6% | ±2.4% |
| XGBoost | 63.2% | 62.3% | 63.2% | 62.7% | 70.3% | ±2.6% |
| Extra Trees | 69.2% | 63.5% | 69.2% | 63.7% | 71.9% | ±1.4% |

### *B. Breast Cancer*

**Table — Breast Cancer Model Comparison (D3)**

| Model | Acc | Prec | Rec | F1 | CV Mean | CV Std |
|---|---|---|---|---|---|---|
| Naive Bayes | 93.0% | 93.0% | 93.0% | 93.0% | 93.0% | ±2.0% |
| Logistic Reg. | 98.2% | 98.2% | 98.2% | 98.2% | 97.4% | ±1.7% |
| KNN | 96.5% | 96.5% | 96.5% | 96.5% | 96.7% | ±1.4% |
| SVM | 98.2% | 98.2% | 98.2% | 98.2% | 97.5% | ±2.0% |
| Decision Tree | 91.2% | 91.6% | 91.2% | 91.3% | 91.0% | ±2.8% |
| Random Forest | 95.6% | 95.6% | 95.6% | 95.6% | 95.6% | ±1.2% |
| XGBoost | 95.6% | 95.7% | 95.6% | 95.6% | 96.5% | ±1.0% |
| Extra Trees | 95.6% | 95.7% | 95.6% | 95.6% | 96.0% | ±1.8% |

## V-F. D2 VS D3 COMPARISON

Table below compares best D2 accuracy (as reported) against D3 test accuracy and 5-fold CV mean. D3 values are consistently lower for tree-based datasets due to overfitting correction, which is the expected and desired behavior.

**Table — D2 vs D3 Performance Comparison**

| Disease | D2 Best Model | D2 Acc | D3 Best Model | D3 Test Acc | D3 CV Mean |
|---|---|---|---|---|---|
| Diabetes | Extra Trees | 84.5% | SVM | 74.7% | 76.0±1.7% |
| Heart Disease | SVM | 92.2% | Naive Bayes | 85.2% | 84.1±4.3% |
| Parkinson's | XGBoost | 92.3% | Random Forest | 92.3% | 88.2±4.8% |
| Liver Disease | Random Forest | 72.0% | Logistic Reg. | 69.2% | 71.9±3.4% |
| Breast Cancer | Random Forest | 95.0% | Logistic Reg. | 98.2% | 97.4±1.7% |

**Fig. 1** — 5-Fold Cross-Validation Accuracy (all models × 5 diseases)

**Fig. 2** — Confusion Matrices, Best Model per Disease

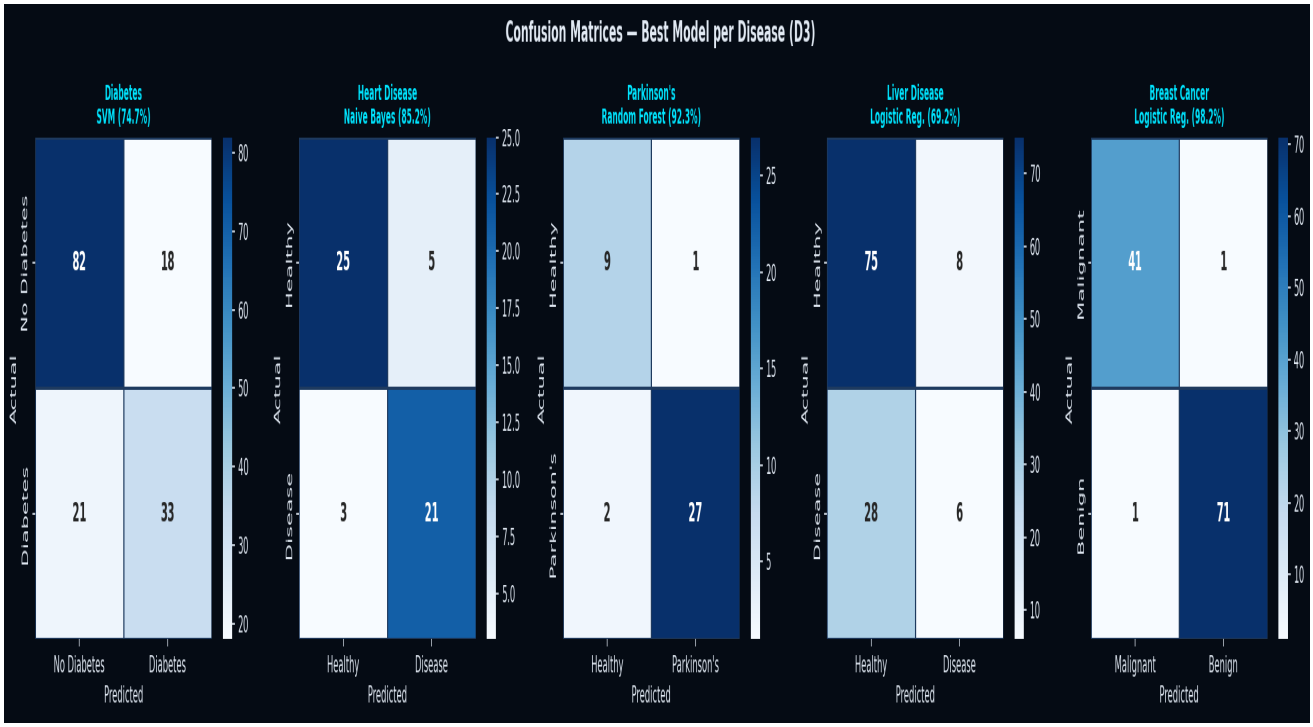



**Fig. 3** — ROC Curves with AUC scores

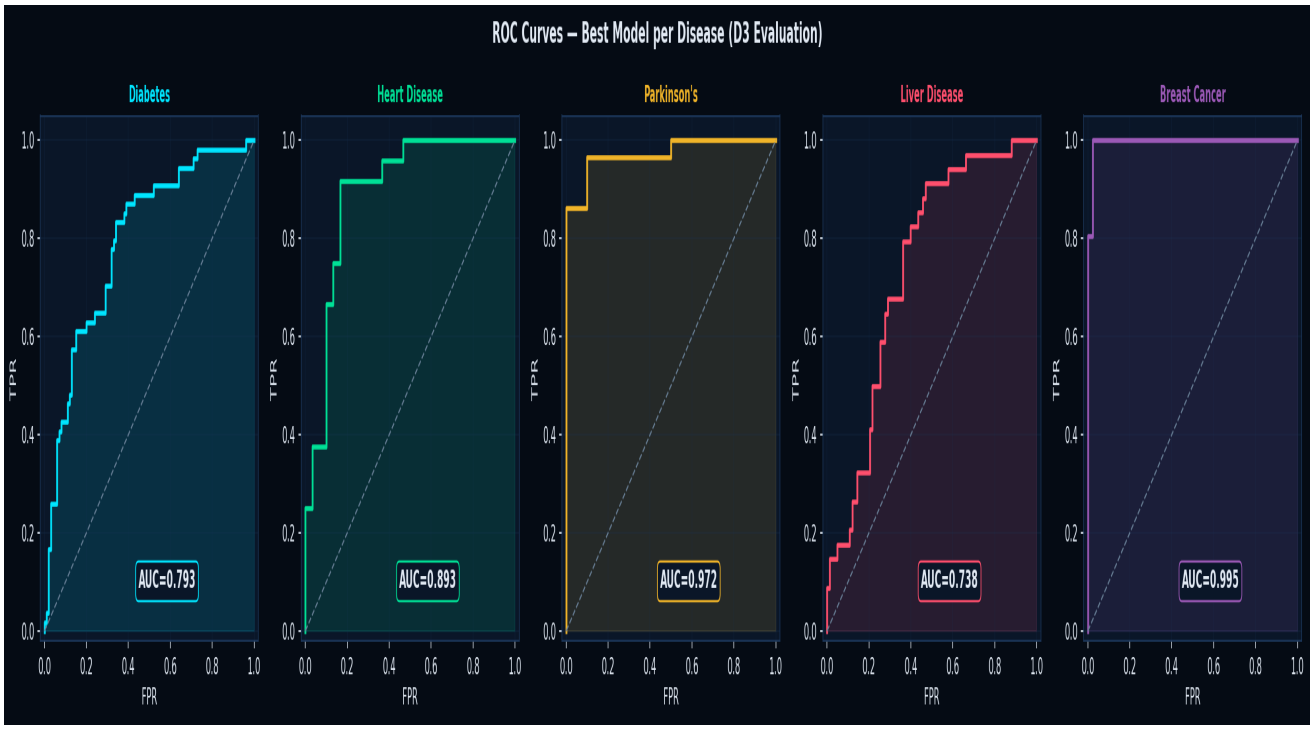



**Fig. 4** — D2 vs D3 Accuracy Comparison

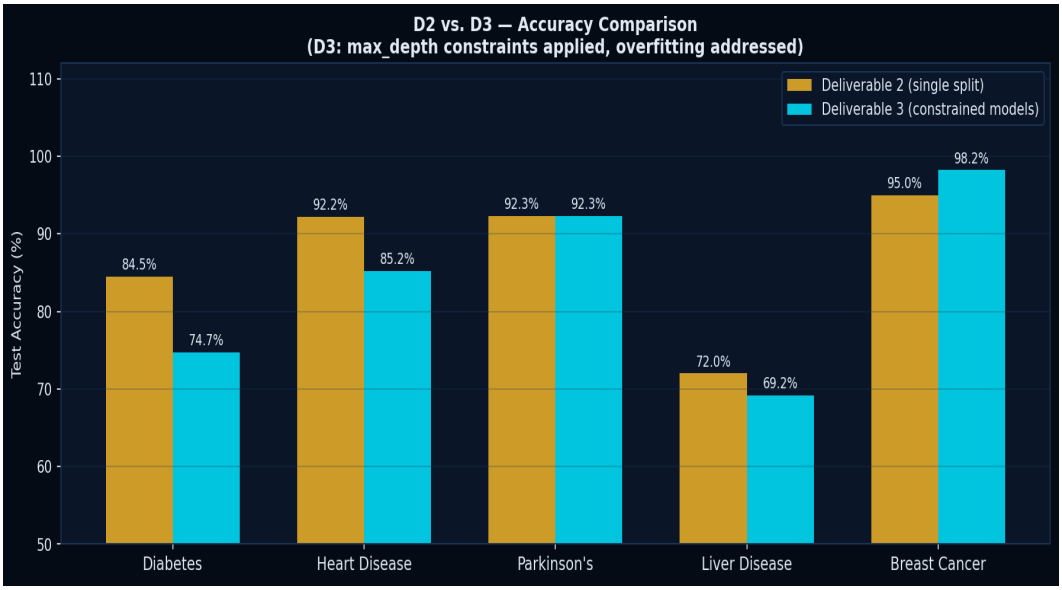



**Fig. 5** — Top-10 Feature Importances (Parkinson's, Liver, Breast Cancer)

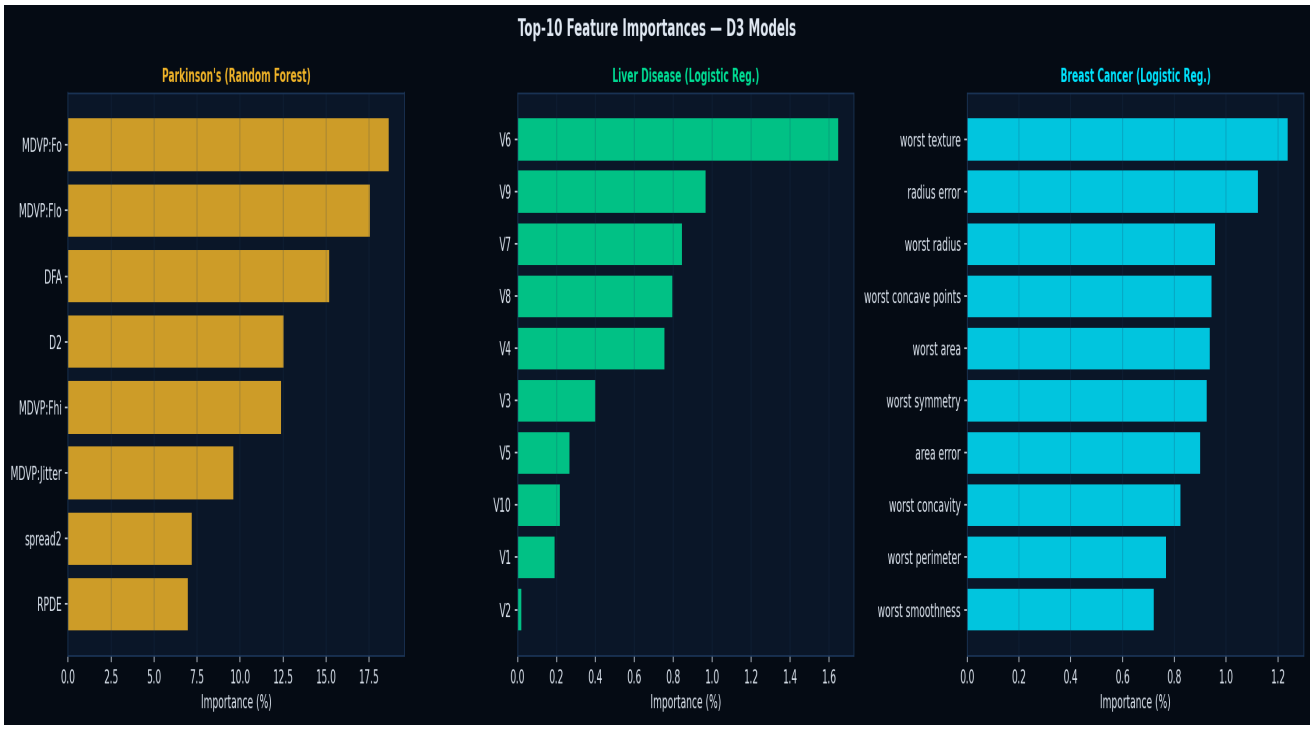
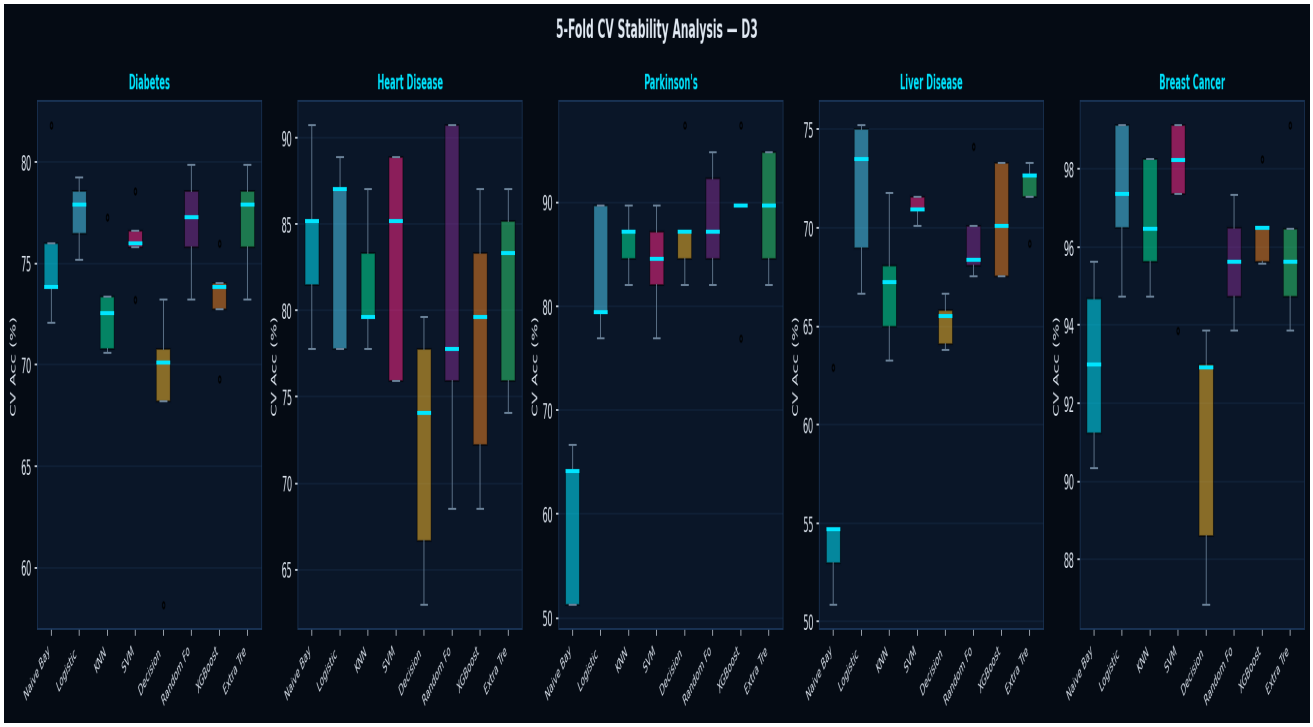



**Fig. 6** — CV Stability Box Plots (5-Fold Score Distribution)




## VI. RESPONSIBLE AI REFLECTION

D3 strengthens responsible AI in three areas identified in D2:

### *A. Transparency and Confidence*

D2 presented binary outputs without probabilities. D3's Doctor Portal surfaces predict_proba() confidence scores color-coded by reliability tier (≥75% high, 55–74% medium, <55% low), giving clinicians an explicit signal of model uncertainty rather than masking it.

### *B. Feature Attribution*

The Doctor Portal's feature importance bars reveal which inputs drove each prediction, enabling a clinician to identify implausible contributions (e.g., a demographic feature dominating a clinical output). This transparency supports audit and challenge of model behavior.

### *C. Fairness and Dataset Limitations*

The PIMA diabetes dataset remains limited to one Native American population (n=768). The liver dataset is drawn from an Indian population cohort (ILPD, n=583). Parkinson's dataset has a strong class imbalance (147 PD vs. 48 healthy). These limitations are now surfaced explicitly in the interface via a disclaimer on each doctor form and in the Important Caveats section of every AI clinical summary.

### *D. Privacy*

No user input is persisted beyond the request lifecycle. The values/ directory used for input logging in development should be disabled or replaced with an anonymized, consent-gated audit log before any clinical deployment.

### *E. Clinical Safety*

The platform adds a mandatory disclaimer on every doctor result: 'AI-generated. For clinical decision support only. Not a substitute for clinical judgment.' The /chat endpoint is instructed to never diagnose and always end medical responses with a reminder to consult a healthcare professional.

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## . REFERENCES


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